Weekly Report 6

Project title: Scot Forge   Date: 03/27/2026

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Progress report: (please indicate the contribution of each of the team members)

We used a probabilistic forecasting approach to estimate gas usage, generating P10, P50, and P95 scenarios to better capture demand uncertainty.

To evaluate performance, we introduced penalty flags as an operational metric, allowing us to compare how different forecasting and decision strategies behave in practice. Using these forecasts, we applied a rolling optimization framework on historical data from 2022–2024 to derive optimal delivery decisions.

In our initial results, the optimized strategy reduced daily penalties by roughly 150 compared to actual operations, while monthly penalties remained about the same (~15). We are continuing to refine the model, particularly around monthly constraint handling.

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Forecasting 2022-2024...
Running optimization...
Evaluating penalties...
Actual               | Daily:  420 | Monthly: 15
Optimized            | Daily:  274 | Monthly: 15
```

From a cost perspective, the optimized deliveries led to approximately 50% fewer Tier 3 cashout charges compared to the actual delivery, which is a promising outcome.

Optimal delivery

Column Labels	1	2	3
Count of Date	799	102	164

Actual Delivery

Column Labels	1	2	3
Count of Date	653	110	302